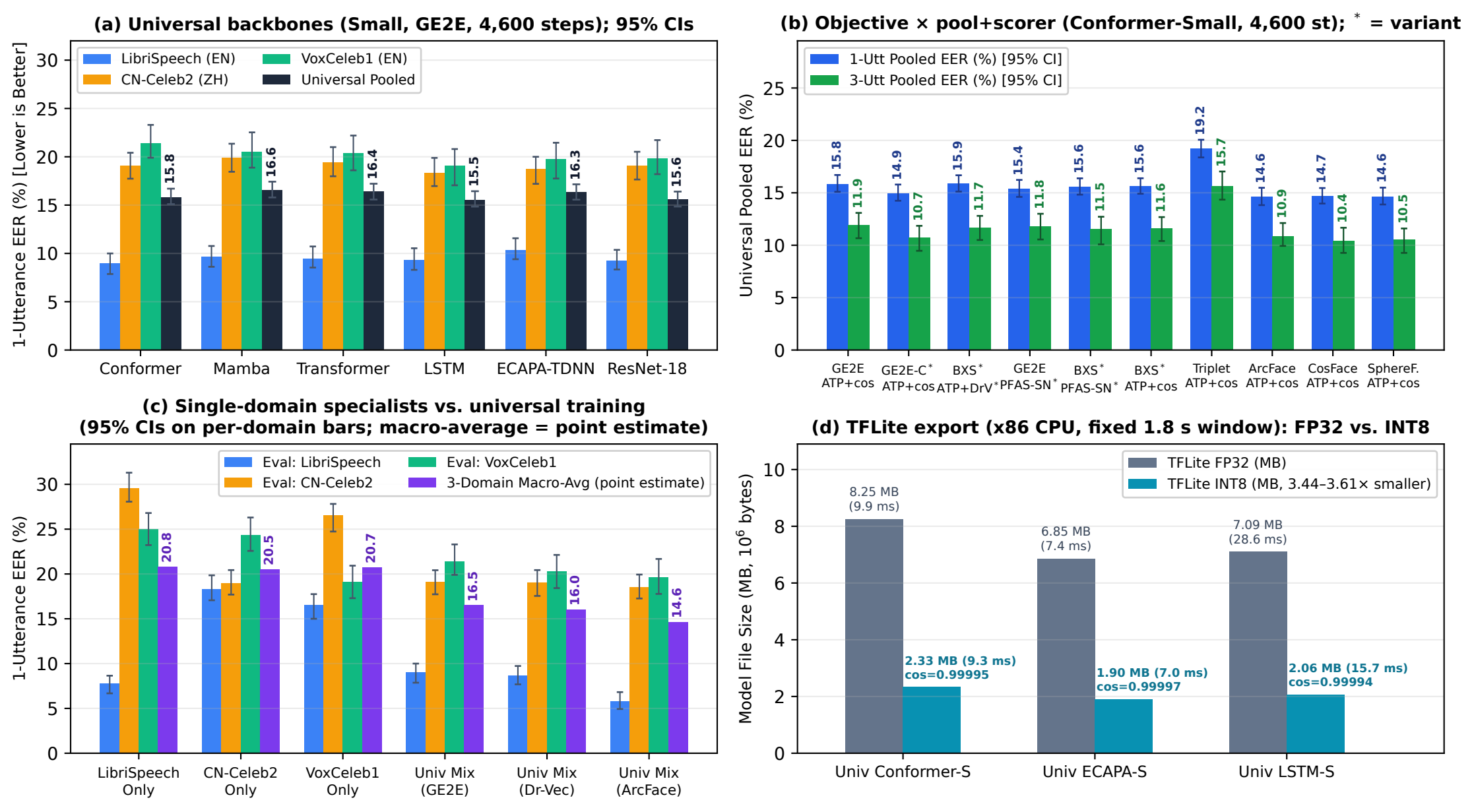
(a) Universal backbones (Small, GE2E, 4,600 steps); 95% CIs
1-Utterance EER (%) [Lower is Better]
LibriSpeech (EN)
CN-Celeb2 (ZH)
VoxCeleb1 (EN)
Universal Pooled
15.8
16.6
16.4
15.5
16.3
15.6
Conformer
Mamba
Transformer
LSTM
ECAPA-TDNN
ResNet-18
(b) Objective × pool+scorer (Conformer-Small, 4,600 st); * = variant
Universal Pooled EER (%)
1-Utt Pooled EER (%) [95% CI]
3-Utt Pooled EER (%) [95% CI]
15.8
11.9
14.9
10.7
15.9
11.7
15.4
11.8
15.6
11.5
15.6
11.6
19.2
15.7
14.6
10.9
14.7
10.4
14.6
10.5
GE2E ATP+cos
GE2E-C* ATP+cos
BXS* ATP+DrV*
GE2E PFAS-SN*
BXS* PFAS-SN*
BXS* ATP+cos
Triplet ATP+cos
ArcFace ATP+cos
CosFace ATP+cos
SphereF. ATP+cos
(c) Single-domain specialists vs. universal training
(95% CIs on per-domain bars; macro-average = point estimate)
1-Utterance EER (%)
Eval: LibriSpeech
Eval: CN-Celeb2
Eval: VoxCeleb1
3-Domain Macro-Avg (point estimate)
20.8
20.5
20.7
16.5
16.0
14.6
LibriSpeech Only
CN-Celeb2 Only
VoxCeleb1 Only
Univ Mix (GE2E)
Univ Mix (Dr-Vec)
Univ Mix (ArcFace)
(d) TFLite export (x86 CPU, fixed 1.8 s window): FP32 vs. INT8
Model File Size (MB, 10^6 bytes)
TFLite FP32 (MB)
TFLite INT8 (MB, 3.44–3.61× smaller)
8.25 MB (9.9 ms)
2.33 MB (9.3 ms) cos=0.99995
6.85 MB (7.4 ms)
1.90 MB (7.0 ms) cos=0.99997
7.09 MB (28.6 ms)
2.06 MB (15.7 ms) cos=0.99994
Univ Conformer-S
Univ ECAPA-S
Univ LSTM-S